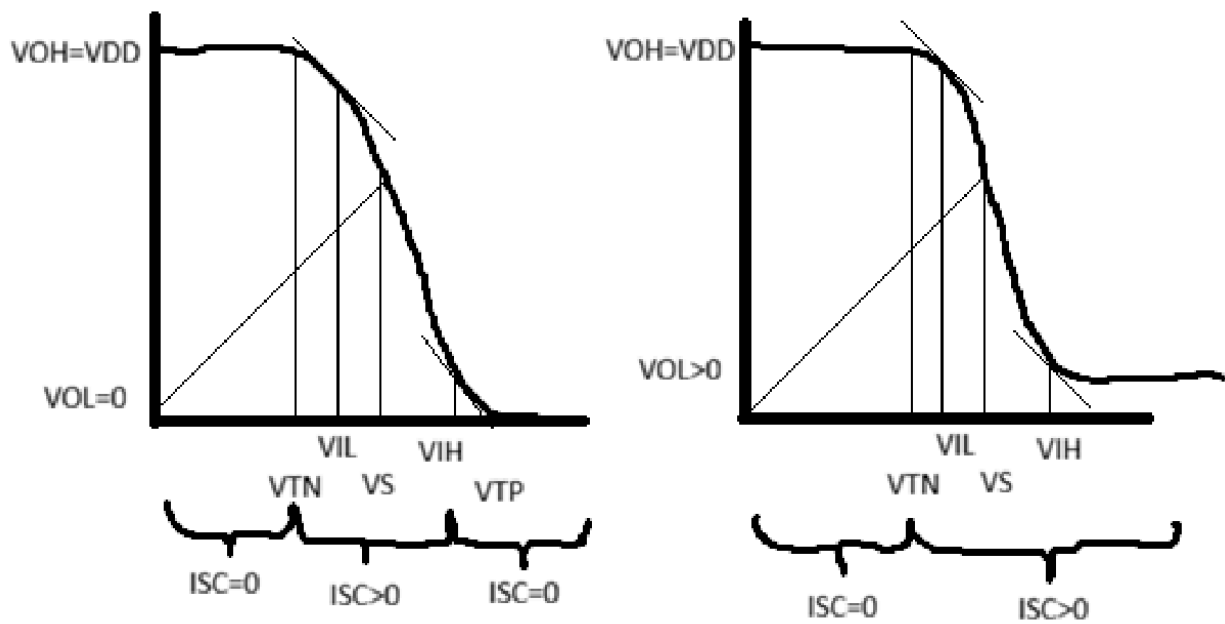


Q1:

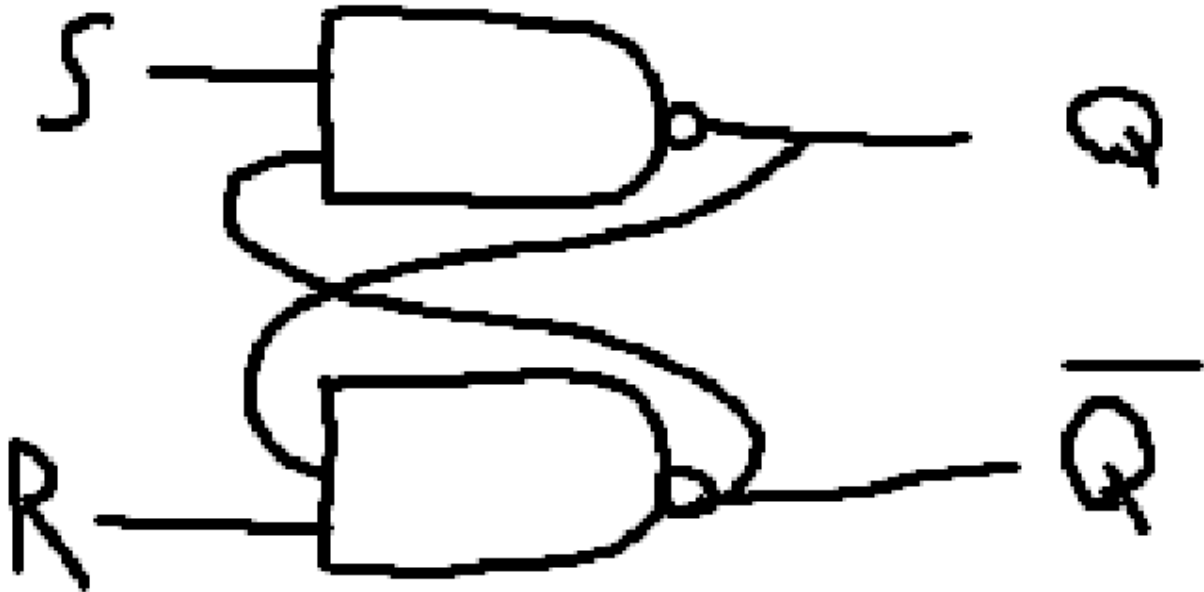
a)



b)

$$PD = \alpha * f * C * VDD^2 = 1 * 2G * 50f * 1.2^2 = 144\mu W$$

Q2:



$$0.7 \cdot R \cdot C_L < 200\text{ps}$$

$$0.7 \cdot 30\text{k} \cdot 100\text{f} / (W_P/L_P) < 200\text{ps}$$

$$W_P/L_P = 10.5$$

for NOR,

$$2(0.7 \cdot 12.5 \cdot 100\text{f} (W_N/L_N)) < 200\text{ps}$$

$$W_N/L_N = 8.75$$